

COVID-19 Early Case Trend Analysis & Recovery

Insights

Project Report

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1. Introduction

COVID-19 is a highly infectious disease that spread rapidly across the world and created significant public health challenges. Analyzing patient-level data helps health authorities understand infection patterns, recovery trends, and regional impacts. This project focuses on analyzing early-stage COVID-19 patient data to generate insights regarding infection spread, patient recovery, and affected regions. The findings can support public health authorities in making data-driven decisions and improving outbreak response strategies.

2. Problem Statement

The public health authority aims to analyze early COVID-19 patient data and answer the following questions:

1. Who is getting infected?
2. How are infections spreading?
3. What are the recovery trends?
4. Which regions are most impacted?
5. What factors influence recovery time?

The objective is to provide meaningful insights that can help improve screening, contact tracing, isolation planning, and public health interventions.

3. Objectives

The main objectives of this project are:

- Perform Exploratory Data Analysis (EDA)
- Analyze demographic patterns of infected patients
- Identify major infection sources
- Evaluate recovery trends
- Compare regional case distributions

- Analyze factors affecting recovery duration
 - Build a Linear Regression model to predict recovery time
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4. Dataset Description

The dataset contains patient-level COVID-19 records with demographic information, infection details, exposure metrics, timeline information, and patient outcomes.

Dataset Summary

- Total Records: 4212
- Total Features: 14

Important Attributes

Attribute	Description
id	Patient ID
sex	Gender
birth_year	Birth Year
country	Country
region	Region
infection_reason	Source of Infection
infection_order	Order of Infection
contact_number	Number of Contacts
confirmed_date	Date of Confirmation
released_date	Date of Recovery
deceased_date	Date of Death
state	Patient Outcome

5. Tools and Technologies Used

Programming Language

- Python

Libraries Used

- Pandas
- NumPy

- Matplotlib
- Seaborn
- Scikit-Learn

Analytical Techniques

- Descriptive Statistics
 - Correlation Analysis
 - Linear Regression
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6. Data Preprocessing

Before analysis, the dataset was cleaned and prepared.

Missing Value Analysis

Several columns contained missing values, particularly:

- birth_year
- infection_reason
- infection_order
- contact_number
- released_date

Missing values were identified and analyzed before performing statistical analysis.

Date Conversion

The following columns were converted into datetime format:

- confirmed_date
- released_date
- deceased_date

Feature Engineering

Age Calculation

Age was calculated using:

Age = 2020 – Birth Year

Recovery Duration

Recovery duration was calculated using:

Recovery Days = Released Date – Confirmed Date

7. Exploratory Data Analysis

7.1 Gender Distribution Analysis

Results

Gender Cases

Female 163

Male 155

Observation

The number of infected male and female patients was almost equal. This indicates that the disease affected both genders similarly in the available dataset.

[Insert Gender Distribution Chart Here]

7.2 Patient Outcome Analysis

Results

Outcome Cases

Isolated 4171

Released 28

Deceased 13

Observation

The majority of patients were still isolated at the time of data collection. Only a small number of patients had recovered or died.

[Insert Patient Outcome Chart Here]

7.3 Country Analysis

Results

Country Cases

Korea 4203

China 8

Mongolia 1

Observation

Most cases were reported in Korea, representing nearly the entire dataset.

[Insert Country Distribution Chart Here]

7.4 Regional Analysis

Top Affected Regions

Region	Cases
Gyeongsangbuk-do	119
Capital Area	109
Daegu	34
Daejeon	10
Gwangju	10

Observation

Gyeongsangbuk-do and Capital Area recorded the highest number of confirmed cases.

[Insert Region Distribution Chart Here]

7.5 Infection Source Analysis

Results

Infection Source	Cases
Contact with Patient	66
Visit to Daegu	40
Visit to Wuhan	8
Pilgrimage to Israel	6

Observation

Direct contact with infected patients was the most common source of infection, highlighting the importance of contact tracing.

[Insert Infection Source Chart Here]

8. Recovery Trend Analysis

Recovery duration was calculated for patients with available recovery dates.

Recovery Statistics

Metric	Value
Mean Recovery Time	15.11 Days
Minimum Recovery Time	7 Days

Metric	Value
Maximum Recovery Time	24 Days
Median Recovery Time	16 Days

Observation

Most recovered patients took approximately 15 days to recover after confirmation.

[Insert Recovery Histogram Here]

9. Correlation Analysis

Correlation analysis was performed to identify relationships between recovery duration and selected variables.

Variable	Correlation with Recovery Days
Age	-0.076
Contact Number	-0.244
Infection Order	0.018

Observation

The correlation values are very close to zero, indicating weak relationships between recovery duration and the selected variables.

[Insert Correlation Heatmap Here]

10. Linear Regression Analysis

A Linear Regression model was built to predict recovery duration.

Model Equation

Recovery Days = 17.03 – 0.0205(Age) – 0.0096(Contact Number) – 0.1912(Infection Order)

Regression Results

Parameter	Value
Intercept	17.03
Age Coefficient	-0.0205
Contact Number Coefficient	-0.0096
Infection Order Coefficient	-0.1912

Interpretation

- Age showed a very weak influence on recovery duration.

- Contact number had a weak negative effect.
- Infection order had a weak negative effect.
- No strong predictor of recovery time was identified.

Limitation

The regression model was trained on a limited number of recovered cases because many records contained missing values.

11. Key Findings

1. Korea accounted for the majority of recorded COVID-19 cases.
 2. Infection distribution was nearly equal among males and females.
 3. Direct patient contact was the primary source of infection.
 4. Gyeongsangbuk-do and Capital Area were the most affected regions.
 5. Average recovery duration was approximately 15 days.
 6. Correlation analysis showed no strong relationship between recovery time and age, contact number, or infection order.
 7. Linear Regression results indicated weak predictive relationships.
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12. Recommendations

- Strengthen contact tracing programs to reduce transmission.
 - Focus preventive measures in high-incidence regions.
 - Improve collection of patient demographic and recovery information.
 - Continue monitoring recovery patterns using larger datasets.
 - Enhance public health surveillance systems for early outbreak detection.
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13. Conclusion

This project successfully analyzed early-stage COVID-19 patient data to understand demographic characteristics, infection patterns, recovery trends, and regional impacts. The analysis revealed that direct contact with infected individuals was the leading infection source and that most cases were concentrated in Korea. Recovery analysis showed an average recovery period of approximately 15 days. Correlation and regression analyses indicated that age, contact exposure, and infection order had limited influence on recovery duration. The insights obtained from this analysis can assist public health authorities in improving outbreak response strategies and resource planning.

14. References

1. COVID-19 Patient Dataset (Provided by Internship Organization)
2. Python Documentation
3. Pandas Documentation
4. Matplotlib Documentation
5. Seaborn Documentation